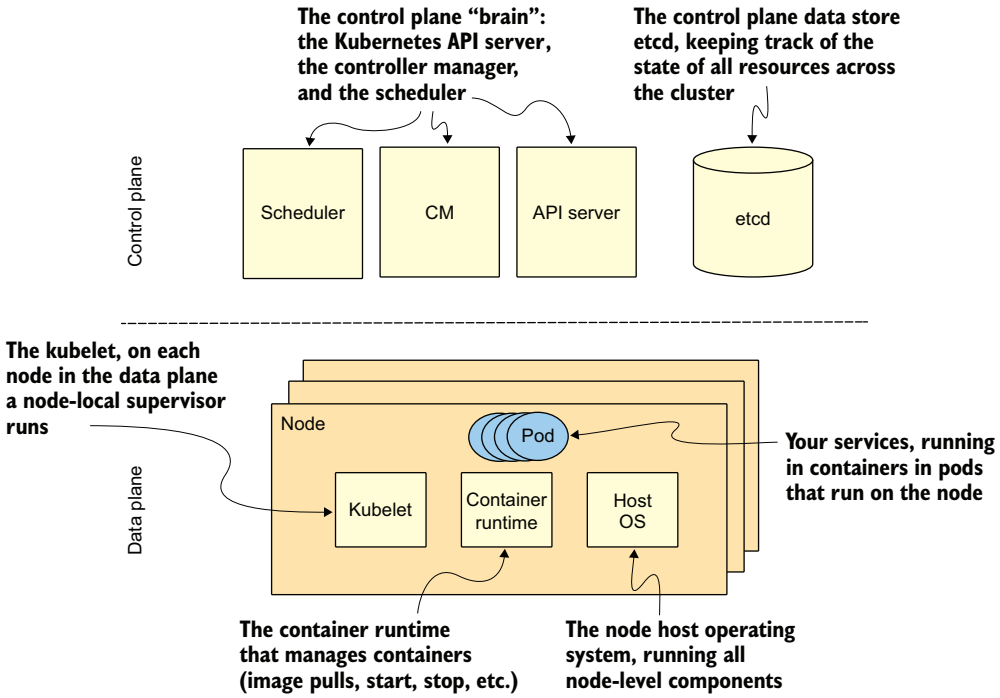


Figure 3.2 Signal sources in Kubernetes

adding support for traces. Let's have a closer look at the different signal types and sources in the Kubernetes context.

Concerning logs in Kubernetes, we can, on a high level, differentiate between the following:

- The system logs (http://mng.bz/EQzq) from the control plane, including the API server, kubelet, and container runtimes.
- The application logs, based on a logging strategy (http://mng.bz/N2DE) you pick (node local or routing based into a central search service, such as OpenSearch).

Depending on the container runtime used, different log management capabilities are available, and as a fallback, you can depend on https://github.com/kubernetes/klog (a Kubernetes-specific Go logging library) being available throughout. Kubernetes doesn't come with a built-in solution for log rotation, either, so this is another decision point, especially concerning portability.

For application logs, Kubernetes supports a simple yet effective way to capture logs that we already know from the section on containers: `stdout` and `stderr`. The transient logs are available via the `kubectl logs` command; however, those are typically not viable for production use, due to their limitations, such as their retention period.